

Ex 3.4

P2. Solution:

$$1. \begin{bmatrix} 2 & 1 & 3 & | & b_1 \\ 6 & 3 & 9 & | & b_2 \\ 4 & 2 & 6 & | & b_3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 3 & | & b_1 \\ 0 & 0 & 0 & | & b_2 - 3b_1 \\ 0 & 0 & 0 & | & b_3 - 2b_1 \end{bmatrix}$$

$$2. Ax=b \text{ is solvable} \Leftrightarrow \begin{cases} b_2 - 3b_1 = 0 \\ b_3 - 2b_1 = 0 \end{cases}$$

3. $C(A)$ is a line spanned by vector $\begin{bmatrix} 2 \\ 6 \\ 4 \end{bmatrix}$ in $\mathbb{R}^3$

4. Special solutions

$$s_1 = \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \end{bmatrix}, s_2 = \begin{bmatrix} -\frac{3}{2} \\ 0 \\ 1 \end{bmatrix}$$

$N(A)$ is the plane in $\mathbb{R}^3$ spanned by

vectors $(-\frac{1}{2}, 1, 0)$ and $(-\frac{3}{2}, 0, 1)$.

$$5. [R \ d] = \begin{bmatrix} 2 & 1 & 3 & | & 10 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \text{ when } b = \begin{bmatrix} 10 \\ 30 \\ 20 \end{bmatrix}$$

Particular solution: set $x_2 = x_3 = 0$, obtain

$$x_p = (5, 0, 0)$$

P4. Solution:

$$[A \ b] = \begin{bmatrix} 1 & 3 & 1 & 2 & | & 2 \\ 2 & 6 & 4 & 8 & | & 8 \\ 0 & 0 & 2 & 4 & | & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 2 & | & 2 \\ 0 & 0 & 2 & 4 & | & 4 \\ 0 & 0 & 2 & 4 & | & 4 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 3 & 1 & 2 & | & 2 \\ 0 & 0 & 2 & 4 & | & 4 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 3 & 0 & 0 & | & \frac{1}{2} \\ 0 & 0 & 1 & 2 & | & \frac{1}{2} \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix} = [R \ d]$$

Particular solution: set $x_2 = x_4 = 0$, solve $Rx = d$.

$$\Rightarrow x_1 = \frac{1}{2}, x_2 = \frac{1}{2}$$

$$\Rightarrow x_p = (\frac{1}{2}, 0, \frac{1}{2}, 0)$$

Special solutions: solve $Rx = 0$

$$s_1 = (-3, 1, 0, 0), s_2 = (0, 0, -2, 1)$$

Complete solution is

$$x = x_p + x_h = \begin{bmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$$P6. 1) [A \ b] = \begin{bmatrix} 1 & 2 & 1 & | & b_1 \\ 2 & 4 & 1 & | & b_2 \\ 2 & 5 & 1 & | & b_3 \\ 3 & 9 & 1 & | & b_4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & | & b_1 \\ 0 & 0 & -1 & | & b_2 - 2b_1 \\ 0 & 1 & -1 & | & b_3 - 2b_1 \\ 0 & 3 & -2 & | & b_4 - 3b_1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 1 & | & b_1 \\ 0 & 1 & -1 & | & b_3 - 2b_1 \\ 0 & 3 & -2 & | & b_4 - 3b_1 \\ 0 & 0 & -1 & | & b_2 - 2b_1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & | & b_1 \\ 0 & 1 & -1 & | & b_3 - 2b_1 \\ 0 & 0 & -1 & | & b_4 - 3b_1 + 3b_1 \\ 0 & 0 & -1 & | & b_2 - 2b_1 \end{bmatrix} = [U \ c]$$

$Ax=b$ is solvable iff

$$\begin{cases} b_4 - 3b_1 + 3b_1 = 0 \\ b_2 - 2b_1 = 0 \end{cases}$$

$$2) [A \ b] = \begin{bmatrix} 1 & 2 & 3 & | & b_1 \\ 2 & 4 & 6 & | & b_2 \\ 2 & 5 & 7 & | & b_3 \\ 3 & 9 & 12 & | & b_4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & | & b_1 \\ 0 & 0 & 0 & | & b_2 - 2b_1 \\ 0 & 1 & 1 & | & b_3 - 2b_1 \\ 0 & 3 & 3 & | & b_4 - 3b_1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 3 & | & b_1 \\ 0 & 1 & 1 & | & b_3 - 2b_1 \\ 0 & 3 & 3 & | & b_4 - 3b_1 \\ 0 & 0 & 0 & | & b_2 - 2b_1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & | & b_1 \\ 0 & 1 & 1 & | & b_3 - 2b_1 \\ 0 & 0 & 0 & | & b_4 - 3b_1 + 3b_1 \\ 0 & 0 & 0 & | & b_2 - 2b_1 \end{bmatrix} = [U \ c]$$

$Ax=b$ is solvable iff

$$\begin{cases} b_4 - 3b_1 + 3b_1 = 0 \\ b_2 - 2b_1 = 0 \end{cases}$$

P8. Solution:

$$(a) C(A) = \mathbb{R}^3$$

$$x_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 6 \\ 3 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{iff } x_1 = x_2 = x_3 = 0$$

$$(b) x_1 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}$$

$$= x_1 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + (x_2 + 2x_3) \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

$\Rightarrow C(A)$ is the plane in $\mathbb{R}^3$ spanned by

vectors $(1, 1, 2)$ and $(1, 2, 4)$.

$$x_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 4 \\ 1 & 4 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

special solution $S = (0, -2, 1)$.

$$\Rightarrow X = c \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

P12. Solution:

$$(a) \begin{cases} AX_1 = b \\ AX_2 = b \end{cases} \Rightarrow \begin{cases} A(X_1 - X_2) = 0 \\ A(X_2 - X_1) = 0 \end{cases}$$

$\Rightarrow S_1 = X_1 - X_2$, and $S_2 = X_2 - X_1$ are two solutions of $AX = 0$

$$(b) A(2X_1 - 2X_2) = 0$$

$$A(X_1 + 2X_1 - 2X_2) = AX_1 + A(2X_1 - 2X_2) = b$$

$\Rightarrow X_1 + 2X_1 - 2X_2 = 3X_1 - 2X_2$ is a solution of $AX = b$.

P25. Solution:

$$(a) r < m, r < n$$

$$(b) r = m < n$$

$$(c) r = n < m$$

$$(d) r = m = n$$

P30. solution:

$$[Ad] \begin{bmatrix} 1 & 0 & 2 & 3 & 1 & 2 \\ 1 & 3 & 2 & 0 & 1 & 5 \\ 2 & 0 & 4 & 9 & 1 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 & 3 & 1 & 2 \\ 0 & 3 & 0 & -3 & 0 & 3 \\ 0 & 0 & 0 & 3 & 0 & 6 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 2 & 0 & 1 & -4 \\ 0 & 3 & 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & 3 & 0 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 & 0 & 1 & -4 \\ 0 & 1 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 & 0 & 2 \end{bmatrix} = [Rd]$$

Particular solution: set $x_3 = 0$, solve $RX = d$

$$\Rightarrow x_1 = -4, x_2 = 3, x_4 = 2$$

$$\Rightarrow X_p = (-4, 3, 0, 2)$$

special solution, set $x_3 = 1$, solve $RX = 0$

$$\Rightarrow x_1 = -2, x_2 = 0, x_4 = 0$$

$$\Rightarrow X_n = (-2, 0, 1, 0)$$

S3.5

P2. Solution: set $A = [V_1, V_2, V_3, V_4, V_5, V_6]$

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & -1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 0 & -1 & -1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & -1 & 0 & -1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

So the largest possible number of independent vectors is 3, such as V_1, V_2, V_3 .

$$P3. i) \text{ If } a = 0, U = \begin{bmatrix} 0 & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix} \rightarrow \text{dependent}$$

$$ii) \text{ If } d = 0, U = \begin{bmatrix} a & b & c \\ 0 & 0 & e \\ 0 & 0 & f \end{bmatrix} \rightarrow \begin{bmatrix} a & b & c \\ 0 & 0 & e \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \text{dependent}$$

$$iii) \text{ If } f = 0, U = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \text{dependent}$$

P6. Solution:

For U :

$$i) \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 9 \\ 0 \end{bmatrix}$$

$$ii) \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 7 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 9 \\ 0 \end{bmatrix}$$

$$iii) \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 7 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 9 \\ 0 \end{bmatrix}$$

For A :

$$i) \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 9 \\ 0 \end{bmatrix}$$

$$ii) \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 7 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 9 \\ 0 \end{bmatrix}$$

$$iii) \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 7 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 9 \\ 0 \end{bmatrix}$$

P8. Proof: $c_1 V_1 + c_2 V_2 + c_3 V_3 = 0$

$$c_1(w_1 + w_3) + c_2(w_1 + w_3) + c_3(w_1 + w_2) = 0$$

$$(c_2 + c_3)w_1 + (c_1 + c_3)w_2 + (c_1 + c_2)w_3 = 0$$

$$\Rightarrow \begin{cases} c_2 + c_3 = 0 \\ c_1 + c_3 = 0 \\ c_1 + c_2 = 0 \end{cases} \Rightarrow c_1 = c_2 = c_3 = 0$$

$\Rightarrow V_1, V_2, V_3$ are linearly independent.

P9. Solution:

(a) $n=4 > m=3$

(b) $V_1 = CV_2$ or $V_2 = CV_1$ for some scalar C .

(c) zero vector is included.

P12. Solution:

$$\underline{AX=b} \quad \underline{A^T y = C}$$

P21. Solution:

i) $\begin{bmatrix} 1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$

special solutions:

$$S_1 = (2, 1, 0), S_2 = (-3, 0, 1)$$

$\{S_1, S_2\}$ forms a basis for the plane.

ii) The intersection of that plane with the xy plane means $\begin{cases} x - 2y + 3z = 0 \\ z = 0 \end{cases}$

$$B = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 0 & 0 \end{bmatrix} \quad BX = 0.$$

special solution: $S = (2, 1, 0)$ which is a basis for the intersection line.

iii) A vector (x, y, z) is perpendicular to that plane, then (x, y, z) is perpendicular to S_1 and S_2 , so that

$$\begin{bmatrix} 2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\begin{bmatrix} 2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 0 \\ 0 & \frac{2}{3} & 1 \end{bmatrix}$$

special solution $S = (\frac{1}{3}, -\frac{2}{3}, 1)$ is a basis.

P23. Solution:

(a) True.

(b) False.

P27. Solution:

(a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

(c) $\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$